

# Structure all the way down: Why mathematics works

Abstract: Why does mathematics work? What are mathematical entities and do they exist in a mind-independent manner? If they exist in a mind-independent manner, how do we come to gain knowledge of them? Mathematical platonists assert that abstract mathematical entities exist mind-independently and that mathematics tracks or constitutes physical structure. The Field-Benacerraf challenge attacks this view on two fronts: Benacerraf's epistemic horn argues that causal inertness makes mathematical entities epistemically inaccessible, whereas Field's metaphysical horn argues that the collapse of the indispensability argument removes the primary empirical motivation for platonism. In this paper, we defend mathematical platonism against both horns by developing an integrated convergence argument. Rather than proceeding in a straightforwardly deductive fashion, our argument functions as an inference to the best explanation. In particular, we identify three independently motivated features of mathematical practice: the constitutive role of mathematics in physical explanation, the repeated convergence of independently developed pure mathematics with physical structure, and the existence of a reliable, non-causal, epistemic route to mathematical knowledge. Mathematical platonism, we argue, provide their best unified explanation. We conclude by drawing the connection between our argument, mathemati-

cal platonism, and ontic structural realism.

Keywords: Mathematical platonism; ontic structural realism; Field-Benacerraf challenge; integrated convergence argument; indispensability argument

## §1 Introduction

Why does mathematics work? This question has two aspects. The first aspect is metaphysical: what are mathematical entities and do they exist in a mind-independent manner?<sup>1</sup> The second is epistemological: if they exist in a mind-independent manner, how do we come to gain knowledge of them? These two aspects motivate the central tension in the philosophy of mathematics – any account worth its salt must provide both a plausible metaphysics and a plausible epistemology, yet these two pull in opposing directions. According to mathematical platonism, abstract mathematical entities exist mind-independently and the success of mathematics in our best scientific theories is not in any way mysterious, since mathematics is tracking something real. The indispensability argument provides support for this core platonist intuition.

Though metaphysically attractive, mathematical platonism struggles to explain how we come to acquire knowledge of these abstract mathematical entities, especially given that they are causally inert and exist outside of spacetime. This is Benacerraf’s epistemic horn. Field sharpens the metaphysical horn at the other end. Mathematical platonists typically believe that our best scientific theories quantify over mathematical entities in ways that cannot be easily paraphrased without loss of explanatory power. If however

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<sup>1</sup> We use ‘mind-independent’ in the ontological sense: entities are mind-independent, just in case they exist even if no minds exist to cognize them. This is distinct from the epistemological question of how we access these mind-independent entities (§ 6).

physics can be nominalized and mathematics thereby rendered dispensable, then the indispensability argument collapses and we lose the primary empirical motivation for mathematical platonism. Benacerraf’s epistemic horn and Field’s metaphysical horn together constitute the Field-Benacerraf challenge.

In this paper, we defend mathematical platonism against the Field-Benacerraf challenge with what we term the integrated convergence argument. Our argument does not attempt to establish mathematical platonism deductively. Instead, it proceeds as an inference to the best explanation: we identify three independently motivated features of mathematical practice and argue that mathematical platonism provides their best unified explanation. These three features correspond to three logically interdependent conditionals:<sup>2</sup>

(First conditional or mathematics as constitutive) If mathematics can be constitutive of physical explanation, then its role in science is not merely representational.

(Second conditional or anti-coincidence) If independent mathematical inquiry repeatedly converges on physical structure, then the fit between mathematics and physical reality is not well explained by mere coincidence.

(Third conditional or structural-inferential knowledge) If mathematical structures can be known without causal interaction, then epistemic access is not a problem.

Our paper proceeds as follows: § 2 reviews mathematical platonism

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<sup>2</sup> The first conditional establishes that mathematics is not merely a representational device (§ 4). The second conditional provides cumulative evidential support for this claim (§ 5), while the third conditional explains how we acquire knowledge of structures that perform this constitutive work (§ 6). Each conditional reinforces and strengthens the others.

and the indispensability argument; § 3 develops the Field-Benacerraf challenge in a more rigorous fashion; §§ 4–6 defend the first, second, and third conditionals. § 7 concludes by drawing the connection between the integrated convergence argument, mathematical platonism, and ontic structural realism.

## §2 Mathematical platonism and the indispensability argument

Mathematical platonism is the metaphysical view that abstract mathematical entities exist in a manner that is independent of us, our language, our thought, and our practices: numbers, functions, and sets exist mind-independently in the way that quarks, electrons, neutron stars, and galaxies do (Frege, 1953; Gödel, 1944, 1947; Maddy, 1990; Plato, 1992). The indispensability argument provides empirical support for mathematical platonism (Colyvan, 1998; Putnam, 1971; Quine, 1948). This argument has its basis in the standard Quinean criterion of ontological commitment:

A theory is committed to those and only those entities to which the bound variables of the theory must be capable of referring in order that the affirmations made in the theory be true. (Quine, 1948, p. 33)

For example, if a theory contains the sentence ‘ $\exists x \text{ Electron}(x)$ ’, then the bound variable  $x$  must range over electrons in order for this theory to be true and this makes the theory ontologically committed to electrons. The Quinean criterion becomes a key premiss in the indispensability argument:

P1 (Quinean criterion): We ought to be ontologically committed to all and only those entities indispensable to our best scientific

theories.

P2: Mathematical entities are indispensable to our best scientific theories (especially physics).

P3/C1: Therefore, we ought to be ontologically committed to mathematical entities.

P4: If we are ontologically committed to  $x$ , then  $x$  exists.

C2: Therefore, mathematical entities exist.<sup>3</sup>

In our best and most successful physical theories, quantum theory uses Hilbert spaces, the movement of planets is described by equations, lawlike structures are described by algebraic functions, and physical quantities are represented by vectors. Indispensability entails that you cannot gut mathematics from physics, without losing explanatory power of why planets follow stable orbits, quantum systems exhibit probabilistic interference patterns, and physical systems obey lawlike regularities. Given that you cannot rewrite physics without mathematics, mathematics is more than mere shorthand, a convenient representational device, or ornament – it is part of the core machinery of physical theories.

At the same time, the indispensability argument can be resisted. Nominalists, for instance, will deny P2, arguing that mathematics is pragmatically convenient rather than genuinely indispensable (Field, 1980). They might

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<sup>3</sup> Let  $Sx$  denote that  $x$  is indispensable to our best scientific theories,  $Ox$  denote that we ought to be ontologically committed to  $x$ ,  $Ex$  denote that  $x$  exists, and  $Mx$  denote that  $x$  is a mathematical entity. Formally:

P1:  $\forall x(Sx \rightarrow Ox)$

P2:  $\exists x(Mx \wedge Sx)$

P3/C1:  $\therefore \exists x(Mx \wedge Ox)$

P4:  $\forall x(Ox \rightarrow Ex)$

C2:  $\therefore \exists x(Mx \wedge Ex)$

This argument is proof-theoretically valid.

attempt to demonstrate how physics can be reconstructed without quantifying over abstract mathematical entities, instead replacing the latter with spatial regions, physical relations, and nominalistic structures. By contrast, fictionalists assert that mathematics is a useful though literally false ‘as-if’ discourse (Balaguer, 1998; Yablo, 2001). Rather than rejecting the Quinean criterion (P1), fictionalists will restrict its application to literal, truth-apt discourse. This allows fictionalists to deny P3 and P4: since mathematical discourse, though useful, is ultimately fictional, it does not genuinely incur any ontological commitment to abstract mathematical entities. Our integrated convergence argument engages with both positions: it challenges nominalism at the level of constitutivity (§ 4) and convergence (§ 5) and fictionalism at the level of epistemology (§ 6).

### §3 The Field-Benacerraf challenge

The Field-Benacerraf challenge provides an important challenge to mathematical platonism (§ 2). This challenge can be represented in the form of the following three premises:

P1: Mathematicians are reliable in the sense that, for (almost) every mathematical sentence  $x$ , if mathematicians accept  $x$ , then  $x$  is true.

P2: If belief in mathematics is to be justified, then it must be possible, at least in principle, to explain the reliability of mathematicians (as described in P1).

P3: If mathematical platonism (as described in § 2) is true, then this reliability cannot be explained, even in principle.

Let  $Ax$  denote that mathematicians accept sentence  $x$ ,  $Tx$  denote that

sentence  $x$  is true,  $R$  denote that mathematicians are reliable,  $E$  denote that belief in mathematics is epistemically justified,  $P$  denote that mathematical platonism is true, and  $Exp$  denote explainability in principle. Formally:

$$\text{P1: } R \leftrightarrow \forall x(Ax \rightarrow Tx)$$

$$\text{P2: } E \rightarrow Exp(R)$$

$$\text{P3: } P \rightarrow \neg Exp(R)$$

Maintaining both  $E$  and  $P$  yields a contradiction:  $Exp(R) \wedge \neg Exp(R)$ . Alternatively,  $E \wedge P \vdash \perp$ . Therefore, if belief in mathematics is to be justified, then mathematical platonism must be false:  $E \rightarrow \neg P$ . P1 and P2 are relatively uncontroversial: most mathematical platonists accept P1 and P2 seems straightforward. P3 is the most controversial premise in the Field-Benacerraf challenge: it has its basis in the observation that the truth values of mathematical assertions depend on facts involving abstract, platonic entities that lie outside of spacetime and are causally isolated from us even in principle (Benacerraf, 1973; Field, 1989). This observation gives rise to a two-pronged attack:

(Benacerraf’s epistemic horn) If mathematical entities are abstract, causally inert, and outside of spacetime, then how can mathematicians even hope to gain reliable epistemic access to these entities?

(Field’s metaphysical horn) If mathematics is dispensable and science can be nominalized, then how can mathematical platonism and belief in the existence of abstract mathematical entities continue to be supported after the collapse of the indispensability argument (§ 2)?

We propose a response on behalf of mathematical platonism – the in-

egrated convergence argument – that can dissolve the epistemic problem and restore the indispensability argument on stronger ground. For convenience, our argument shall be termed the integrated convergence argument. We propose to do so by defending the truth of three logically interdependent conditionals:

(First conditional or mathematics as constitutive) If mathematics can be constitutive of physical explanation, then its role in science is not merely representational.

(Second conditional or anti-coincidence) If independent mathematical inquiry repeatedly converges on physical structure, then the fit between mathematics and physical reality is not well explained by mere coincidence.

(Third conditional or structural-inferential knowledge) If mathematical structures can be known without causal interaction, then epistemic access is not a problem.

The cumulative force of all three conditionals, we argue, decisively favours mathematical platonism over its philosophical rivals.

#### **§4 The first conditional or mathematics as constitutive**

Our first conditional claims that mathematics can be constitutive of physical explanation. Some precision about what this claim requires and does not require will be useful. This claim entails neither that every physical fact is identical to a mathematical fact nor that any nominalistic paraphrase is logically impossible. Rather, the claim is more restricted in its scope:

(Constitutivity') In certain cases, the explanatory work performed

by a physical theory cannot be preserved under nominalistic paraphrase without loss of explanatory power.

The relevant issue therefore is not whether a nominalistic paraphrase is logically possible but whether it preserves modal structure, counterfactual coherence, explanatory dependence, and so on. While Field's (1980) nominalization programme could preserve the truth conditions for a number of physical claims, it may not preserve explanatory structure in the relevant sense. Our constitutivity' claim targets this gap between the preservation of truth conditions and the preservation of full explanatory power. The mathematical ground floor consists of measure-theoretic structures. Non-measurable sets – Vitali sets and Banach-Tarski constructions – are not speculative conjectures but rather derived consequences of ZFC set theory (containing the Axiom of Choice) will be committed to (Banach and Tarski, 1924; Fraenkel, 1922; Fraenkel, Bar-Hillel, and Levy, 1973; Vitali, 1905; Zermelo, 1904, 1908). Their existence renders naive measure theory inconsistent and the domain of measure must be restricted to a  $\sigma$ -algebra satisfy closure under certain mathematical constraints. One of these constraints is countable additivity, according to which for any countable collection of disjoint sets  $A_1, A_2, A_3, \dots$  belonging to  $\Sigma$ :<sup>4</sup>

$$\text{(Countable additivity)} \quad \mu(A_1 \cup A_2 \cup A_3 \cup \dots) = \mu(A_1) + \mu(A_2) + \mu(A_3) + \dots$$

Another of these constraints is measure zero:

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<sup>4</sup>  $\Sigma$  is a collection of subsets of  $X$ , given an underlying set  $X$ .  $\Sigma$  marks the boundary between sets that a measure can assign a size to (measurable) and sets that it cannot (non-measurable). Without  $\Sigma$ , a measure would have to assign a size to every subset of  $X$ . Vitali sets and Banach-Tarski constructions demonstrate that this would be impossible under ZFC theory.

(Measure zero)  $\mu(\emptyset) = 0$

The restricted domain  $\Sigma$  is a  $\sigma$ -algebra. The measure, in turn, is a function  $\mu : \Sigma \rightarrow [0, \infty]$  on this  $\sigma$ -algebra, satisfying the constraints of countable additivity and measure zero. The abstract measure space is a forced minimal structure that can be represented by the triple  $(X, \Sigma, \mu)$ , with  $X$  denoting the underlying set,  $\Sigma$  denoting the  $\sigma$ -algebra on  $X$ , and  $\mu$  denoting the measure function. The probability space is the measure space with an additional constraint:

(Normalization)  $\mu(X) = 1$

The probability space can therefore be represented by the triple  $(X, \Sigma, \mu)$ , with  $\mu(X) = 1$ . Where  $\Omega$  denotes the sample space,  $\mathcal{F}$  denotes the  $\sigma$ -algebra of events, and  $P$  denotes the probability measure, the probability space can equally be represented by the triple  $(\Omega, \mathcal{F}, P)$ , with  $P(\Omega) = 1$ . The same structures that non-measurable sets forced into existence – the  $\sigma$ -algebra, the probability measure, the Hilbert space, and the spectral theorem – turn out to be exactly the structures required by quantum mechanics. These mathematical structures constitute what it means for a probability space to be well-defined and what it means for quantum-mechanical observables to have determinate, real-numbered values.

This supports constitutivity' in a strong form. The non-commutativity of operators on a Hilbert space, it may be argued, encodes modal constraints governing possible measurement outcomes, rather than merely representing the impossibility of simultaneously measuring variables such as momentum and position. A nominalistic paraphrase would have to reproduce this modal structure, without quantifying over any mathematical objects. No such nominalistic reformation has been achieved, and this explanatory gap reflects the

constitutive role of the Hilbert-space formalism in defining the nature of quantum-mechanical observables.

Noether's (1918) theorem extends constitutivity' beyond quantum mechanics and to the physical laws themselves. Conservation of energy, momentum, and angular momentum are not empirical generalizations that are subsequently described in mathematical language. Instead, they consist of mathematical theorems about the continuous symmetries of physical systems.<sup>5</sup> The conservation laws are not merely described or represented by mathematics; they are the mathematics, under physical instantiation.

Through Noether's theorem, constitutivity' applies to classical and relativistic mechanics, broadening the scope of our first conditional. Nominalists might object that Noether's theorem can be restated in nominalistic terms, using spacetime regions and physical fields. Constitutivity' does not require us to deny that this theorem has a nominalistic correlate. Rather, our claim is that such a nominalistic correlate would be preserve explanatory structure: it would fail to explain why the continuous symmetry generates a conserved quantity. Such an explanation, it may be argued, can only be constituted by the mathematical proof.

## **§5 The second conditional or anti-coincidence**

According to the first conditional, if mathematics can be constitutive of physical explanation, then its role in science is not merely representational (§ 4). Our second conditional holds that repeated, independent convergence be-

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<sup>5</sup> To be precise, the conservation of energy corresponds to time-translation symmetry, the conservation of momentum corresponds to spatial-translation symmetry, and the conservation of angular momentum corresponds to rotational symmetry (Noether, 1918).

tween pure mathematics and physical reality is not well explained by an appeal to coincidence. The crucial argumentative move involves not merely noting that such convergences occur: following Eugene Wigner's (1960) point about the unreasonable effectiveness of mathematics, this is already widely acknowledged. Rather, we will follow up with a question about which hypothesis best explains this repeated pattern of independent convergences.

A distinction can be made between three explanatory hypotheses:

( $H_0$  or deflationary coincidence) The convergences are brute-fact coincidences and do not admit of any deeper explanation.

( $H_1$  or nominalism/fictionalism) Mathematics is a representational tool and successful applications of pure mathematics in the physical domain result from iterative selection and pragmatic adaptation to the explanatory needs of physics.

( $H_2$  or platonism) Mathematical structures exist mind-independently and successful applications of pure mathematics in the physical domain result from the instantiation or approximation of mathematical structures in physical reality.

The convergence data  $E$  consists of examples of convergence that satisfy three conditions:

(C1 or independence) The pure mathematics was developed independently of any empirical motivation and under purely internal constraints;

(C2 or applicability) The pure-mathematical development later proved applicable in a physical domain with a high degree of precision;

(C3 or scope) These convergences span distinct mathematical and physical domains.

Under  $H_0$ , each such convergence is a brute fact, but these brute facts cannot multiply with explanatory cost. Under  $H_1$ , convergences satisfying C1 (independence) are unexpected, since selection and adaptation can account for the mathematics developed alongside or after the physical theory, but not for the mathematics that precedes and constrains the physical theory. Under  $H_2$ , these convergences are expected: independently developed pure mathematics (C1) will sometimes match physical structures (C2), even across multiple, distinct mathematical and physical domains (C3), because physical reality instantiates certain mathematical structures that are consistent with the laws of physics. According to Bayes' rule, as the number  $n$  of independent convergences satisfying C1–C3 grows, we expect that  $P(E \mid H_0)$  and  $P(E \mid H_1)$  will decrease multiplicatively, whereas  $P(E \mid H_2)$  should not. In fact, the posterior odds should shift decisively in favour of  $H_2$ .

Two convergences should suffice to illustrate our argument. The first convergence involves Lie group theory and Noether's theorem (Lie and Engel, 1888–1893; Noether, 1918). The mathematics of continuous symmetries – Lie group theory – was first developed as a purely mathematical programme in the late 1800s, with no physical motivation. Emmy Noether applied it to variational problems in the 1910s. The physical significance of Noether's theorem – that mathematics is constitutive of all conservation laws in physics – was absorbed by physicists only gradually, long after the mathematics had been derived under purely internal constraints. C1 is clearly satisfied here: the pure mathematics preceded and was independent of its physical application.

The second convergence involves Ramanujan's summation and the Casimir

effect (Berndt, 1985; Casimir, 1948). It is even more striking and dramatic than the first convergence, because the pure-mathematical result is, at surface level, physically absurd. According to Ramanujan's summation:

$$1 + 2 + 3 + 4 + 5 + \cdots = -\frac{1}{12} \text{ (Berndt, 1985, p. 135)}$$

The divergent series  $1 + 2 + 3 + 4 + 5 + \cdots$  does not converge in the ordinary sense, since its partial sum grows without bound. Assigning it a finite, negative value therefore appears nonsensical from any physically motivated standpoint. Working in isolation in the early 1900s in Madras and well before his correspondence with Hardy began in the 1910s, Ramanujan derived this result through the analytic continuation of the Riemann zeta function ( $\zeta(-1) = -\frac{1}{12}$ ) with no empirical motivation. Ramanujan's result satisfies C1 in the strongest possible sense

An informal derivation of Ramanujan's summation follows the standard heuristic path of the Grandi series and Cesàro summation:<sup>6</sup>

1.  $S = 1 + 2 + 3 + 4 + 5 + \cdots$
2.  $S_1 = \sum_{n=0}^{\infty} (-1)^n = 1 - 1 + 1 - 1 + 1 - \cdots$  (this is also known as a Grandi (1710) series)
3.  $1 - S_1 = 1 - (1 - 1 + 1 - 1 + 1 - \cdots) = 1 - 1 + 1 - 1 + 1 - \cdots = S_1$   
(from 2)
4.  $\therefore 2S_1 = 1$  (from 3)
5.  $\therefore S_1 = \frac{1}{2}$  (from 4; this is also known as a Cesàro (1888) summation)

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<sup>6</sup> The more rigorous route relies on the analytic continuation of the Riemann zeta function. While our heuristic derivation is presented for expository clarity, the physical application of Ramanujan's summation relies on zeta-regularization.

6.  $S_2 = \sum_{n=1}^{\infty} (-1)^{n+1} n = 1 - 2 + 3 - 4 + 5 - \dots$
7.  $S_1 - S_2 = (1 - 1 + 1 - 1 + 1 - \dots) - (1 - 2 + 3 - 4 + 5 - \dots) =$   
 $(1 - 1) + (-1 + 2) + (1 - 3) + (-1 + 4) + (1 - 5) + \dots =$   
 $1 - 2 + 3 - 4 + 5 - \dots = S_2$  (from 2 and 6)
8.  $\therefore S_1 = 2S_2$  (from 7)
9.  $\therefore S_2 = \frac{1}{4}$  (from 5 and 8)
10.  $S - S_2 = (1 + 2 + 3 + 4 + 5 + \dots) - (1 - 2 + 3 - 4 + 5 - \dots) =$   
 $4 + 8 + 12 + \dots = 4(1 + 2 + 3 + \dots) = 4S$  (from 1 and 6)
11.  $\therefore -S_2 = 3S$  (from 10)
12.  $\therefore S = -\frac{1}{4} \times \frac{1}{3} = -\frac{1}{12}$

The Casimir (1948) effect provides a physical realization of this pure-mathematical result. According to this effect, two uncharged, perfectly conducting parallel plates in vacuum will attract each other. The measurable attractive force can be computed from the change in zero-point energy of the field when the plates are brought into position. In a simplified, one-dimensional field, the zero-point energy involves  $\sum_{n=1}^{\infty} n$ , which diverges. Zeta regularization, however, assigns  $\zeta(-1) = -\frac{1}{12}$  to this sum, yielding a finite, negative energy that represents plate attraction. The Casimir effect has been experimentally confirmed to a high degree of precision (Lamoreaux, 1997). All things considered, the same pure-mathematical result that appeared physically absurd has turned out to constitute a measurable, experimentally verified attractive force in the Casimir effect.

The two convergences – Lie group theory and Noether’s theorem and Ramanujan’s summation and the Casimir effect – satisfy C1–C3 in the relevant sense: Lie group theory and Ramanujan’s summation were developed

independently of any empirical motivation and under purely internal constraints (C1), are drawn from distinct mathematical traditions such as continuous symmetry theory and analytic number theory (C3), later proved applicable (C2) to distinct physical domains (C3). Under  $H_0$  (deflationary coincidence) and  $H_1$  (nominalism/fictinalism), an independent explanation is required for each convergence. Under  $H_2$  (platonism), by contrast, both are instances of the same phenomenon: physical reality instantiating certain mathematical structures whose properties are accessible by structural-inferential means (§ 6).

## §6 The third conditional or structural-inferential knowledge

According to the first conditional, if mathematics can be constitutive of physical explanation, then its role in science is not merely representational (§ 4). According to the second conditional, if independent mathematical inquiry repeatedly converges on physical structure, then the fit between mathematics and physical reality is not well explained by mere coincidence (§ 5). Our third conditional holds that mathematical structures can be known without causal interaction. Therefore, epistemic access turns out not to be the problem that Benacerraf's epistemic horn supposes it to be (§ 3). We defend this third conditional as an abductive thesis: the reliability of structural inference stands in need of an explanation and mathematical platonism provides the best unified account. Benacerraf's epistemic horn (§ 3) implicitly assumes that causal-perceptual contact is the sole epistemic route to knowledge of an entity. We resist this assumption by distinguishing between at least two epistemic routes:<sup>7</sup>

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<sup>7</sup> Epistemic pluralism of this sort has been defended elsewhere by Resnik (1997) and Shapiro (1997). Our version, however, differs from theirs in how we tie the reliability of

(Causal-perceptual knowledge) Direct contact with physical objects, the preferred epistemic route of nominalism;  
 (Structural-inferential knowledge) Knowledge derived by following the forced consequences of a consistent axiomatic system (§ 3).

Structural-inferential knowledge should not be confused with mere formal manipulation. Its reliability conditions include internal consistency, inferential stability, and (in certain instances) external success in application. If independently derived mathematical structures repeatedly prove applicable in describing physical reality, then structural inference would appear to be tracking structure that is also instantiated in the physical world (§ 5). The history of mathematics exhibits a systematic pattern of structural-inferential extension: established mathematical frameworks have been extended through the relaxation of constraints that had previously seemed necessary, yielding richer structures that remain internally consistent. Three cases are instructive:

(Real numbers to complex numbers) The constraint dropped was closure under real solutions or the requirement that polynomial equations had to have real-valued roots. Admitting  $\sqrt{-1}$  as a legitimate mathematical object yields a richer, consistent algebraic structure that has a domain of application in quantum mechanics and signal processing;  
 (Euclidean to non-Euclidean geometry) The constraint dropped was the parallel postulate or the requirement that through any point not on a line, exactly one parallel line can be plotted. Dropping this postulate yields hyperbolic and elliptic geometries that

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structural-inferential knowledge explicitly to our convergence data  $E$  in § 5, making the epistemological and metaphysical aspects of our argument interdependent and mutually supporting rather than independent and isolated.

are consistent and – as it turns out – physically realized in general relativity. However, the pure-mathematical derivation preceded its physical application by several decades (§ 5); (Naive to non-measurable set theory) Dropping the constraint that every subset of a set  $X$  can be assigned a measure and accepting the Axiom of Choice forces non-measurable sets (Vitali sets and Banach-Tarski constructions) into existence as derived consequences rather than mere stipulations. In addition, these structural-inferential moves lead to the generation of the  $\sigma$ -algebra and measure space that we have already used in support of our constitutivity' claim (§ 4).

In each case, the structural-inferential extension was driven by internal mathematical pressure rather than external causal contact. The nominalist demand for causal contact as a condition for all knowledge overextends a principle intended for causal-perceptual knowledge into a domain where structural-inferential knowledge operates under different though equally legitimate reliability conditions. Our third conditional is not independent of the second conditional. The reliability of structural-inferential knowledge in mathematics – the fact that it systematically converges with the structure of physical reality – is itself in need of an explanation. We now have the resources to provide an abductive explanation:

P1: If structural inference were unconstrained by mind-independent reality, then its convergence with physical structure would be coincidental.

P2: If this convergence is systematic and cross-domain (see § 5), then the best explanation is that structural inference tracks structural features shared by both mathematical and physical domains.

C: Therefore, the reliability of structural-inferential knowledge is best explained by the mind-independence of the structures it tracks.

Our argument here is abductive rather than deductive: it does not prove that mathematical structures are mind-independent. Rather, it establishes that mind-independence provides the best unified account of the reliability of structural inference.<sup>8</sup> Instead of appealing to some mysterious causal contact with abstract mathematical objects, we have shown that the reliability and success of structural-inferential knowledge calls for an explanation that mathematical platonism is uniquely poised to provide.

## §7 Conclusion

In conclusion, we have defended an integrated convergence argument for mathematical platonism against the Field-Benacerraf challenge. Our argument involves three logically interdependent conditionals: the first (mathematics as constitutive) establishes that mathematics plays a constitutive rather than merely representational role in physical explanation; the second (anti-coincidence) provides cumulative evidential support for the first through a Bayesian competing-hypothesis analysis of the convergence between pure mathematics and physical reality; and the third (structural-inferential knowledge) allows us to show abductively that mathematical knowledge requires neither causal contact nor some mysterious Platonic access but rather a structural-inferential epistemic route.

We do not claim to have provided a conclusive defence of mathematical

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<sup>8</sup> Here, ‘best’ is evaluated in terms of the standard criteria of scope, simplicity, and the absence of ad-hocness.

platonism. Each of our three conditionals is individually defeasible. Taken together, however, they form a mutually reinforcing explanatory package whose cumulative force exceeds that of its individual components. Our claim is comparative in nature: mathematical platonism provides a better unified account of constitutivity' (first conditional), convergence (second), and reliable structural inference (third) than nominalism or fictionalism: it is better in scope (platonism explains all three features), simplicity (platonism requires only a single ontological posit), and absence of ad-hocness (it does not require each independent convergence to be explained separately).

A further connection emerges between our integrated convergence argument and ontic structural realism (Costa and French, 2003; French, 1999, 2014; French and Ladyman, 2003a,b; Ladyman and Ross, 2007). According to the latter view, objects in the mind-independent world can be reconceptualized as structures.<sup>9</sup> Mathematical platonism tells us that mathematics includes mind-independent structures that are causally inert and exist outside of spacetime (complex numbers, non-measurable sets, and so on). They exist, insofar that knowledge of these structures can be derived by following the forced consequences of a consistent axiomatic system (§ 6). Ontic structural realism tells us what physics captures: physical structures or physically instantiated mathematical structures that satisfy nomological constraints.

The upshot is neither that every mathematical structure has a physical application nor that physical reality reduces to mathematics. Rather, it is that the best explanation of mathematical knowledge – and of the ap-

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<sup>9</sup> French and Ladyman (2003a,b) distinguish epistemic structural realism (only knowledge of structural relations is possible) from ontic structural realism (structure is all there is). Our argument aligns more naturally with ontic structural realism: if physical structures are mathematically constituted or (equivalently) at least some mathematical structures are physically instantiated, then the object/structure distinction dissolves at the level of fundamental physics.

plicability of at least some of this knowledge to physical reality – is that both consist in knowledge of structure, which we can access epistemically by either structural-inferential or causal-perceptual means. Mathematical platonism and ontic structural realism are therefore not merely compatible: under the integrated convergence argument, each provides support for the other. Structure, as it turns out, goes all the way down.

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